

The Emissions of Our Electricity Sources

How many greenhouse gases are produced in each source's lifecycle

Electricity looks identical at the wall socket; the carbon cost of producing it does not. Operational emissions - the chimney number - undercount fossil fuels and overcount renewables. Lifecycle emissions, measured from mine to decommissioning, are the right ruler. On that ruler, nuclear emits 5.6 grams of CO₂-equivalent per kilowatt-hour. Coal emits 970 - a 173× gap. Wind, solar, hydropower, and nuclear all sit below 120; gas, oil, and coal start above 400. Public argument treats nuclear as a fossil-fuel cousin. By the only measure that compares plants on the same terms, it isn't.

Legend

Each form is one electricity source. Area is proportional to lifecycle emissions, in grams of CO₂-equivalent per kilowatt-hour delivered. Green marks nuclear; grey marks other sources.

- Green marks nuclear
- Grey marks other sources

Two reference forms shown below at actual size: nuclear at 5.6 gCO₂/kWh and coal at 970. The ratio between every pair of forms on the poster is shown to the same scale.



Methodology

Median values from peer-reviewed lifecycle assessments compiled by Our World in Data, IPCC AR5, and UNECE. Real projects vary by site and fuel grade - ranges overlap at the edges, not the centre.

Data: Our World in Data (2024), drawing on IPCC AR5 WG3 Annex III and UNECE lifecycle assessment literature. Visualisation: Court Granville, 2026.

